COMP7903 Mobile Forensic Lab

DDL: 27th March (Fri) 23:59PM

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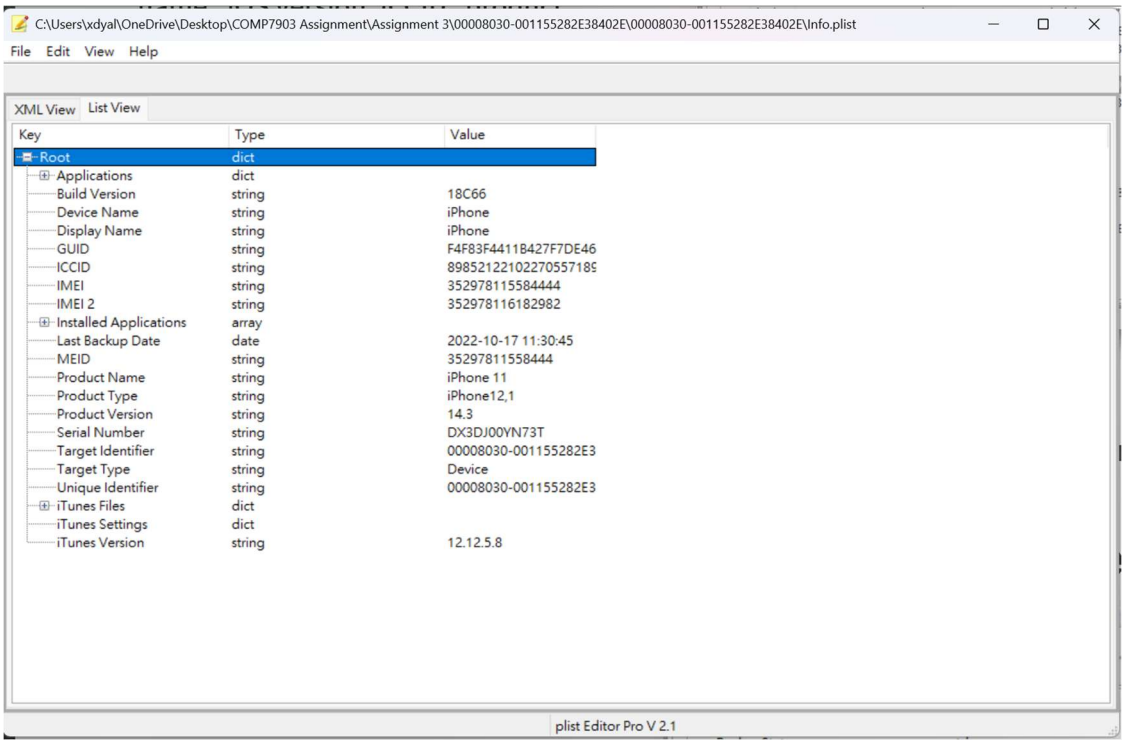
Case1: Please analyze the iTunes Backup file "00008030-001155282E38402E.zip", and answer following questions with screen capture and your analysis detail. (50')

1. <u>What is the IMEI number for this iPhone?</u>

Answer:
IMEI: 352978115584444
IMEI 2: 352978116182982

The IMEI number can be found in the info.plist file.
Please refer to the screen capture below.



Key	Type	Value
Root	dict	
Applications	dict	
Build Version	string	18C66
Device Name	string	iPhone
Display Name	string	iPhone
GUID	string	F4F83F4411B427F7DE46
ICCID	string	8985212210227055718
IMEI	string	352978115584444
IMEI 2	string	352978116182982
Installed Applications	array	
Last Backup Date	date	2022-10-17 11:30:45
MEID	string	35297811558444
Product Name	string	iPhone 11
Product Type	string	iPhone12,1
Product Version	string	14.3
Serial Number	string	DX3DJ00YN73T
Target Identifier	string	00008030-001155282E3
Target Type	string	Device
Unique Identifier	string	00008030-001155282E3
iTunes Files	dict	
iTunes Settings	dict	
iTunes Version	string	12.12.5.8

2. <u>What instant messaging apps are installed on your phone?</u>

A: WhatsApp

B: WeChat

C: LINE

D: Signal

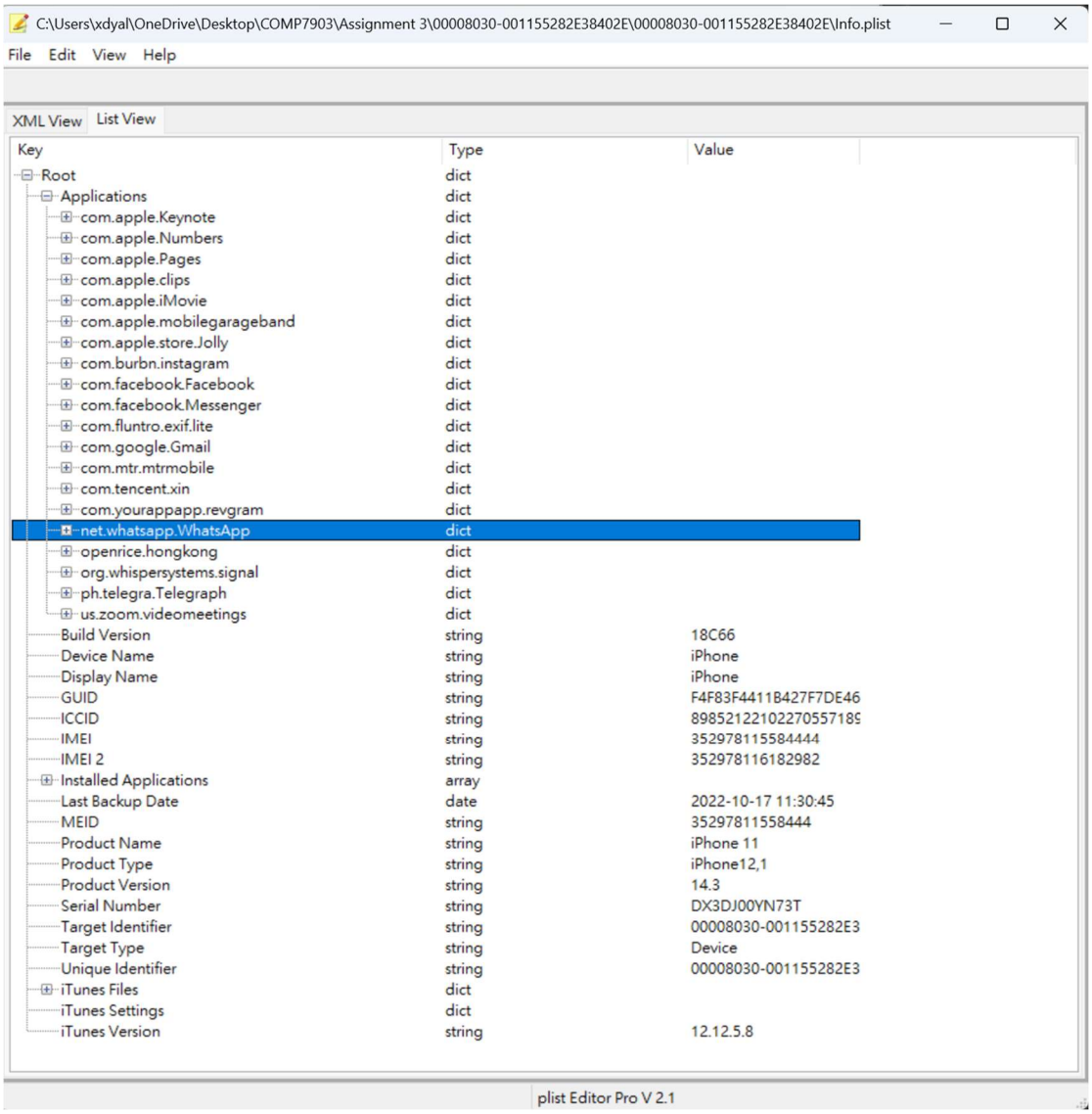
E: QQ

Answer: WhatsApp, WeChat, and Signal

After opening the "Info.plist" file using the plist editor, the installed application information can be found in the "Application" part.

- net.whatsapp.WhatsApp
- com.tencent.xin
- org.whispersystems.signal

The screen capture is shown below:



Key	Type	Value
Root	dict	
Applications	dict	
com.apple.Keynote	dict	
com.apple.Numbers	dict	
com.apple.Pages	dict	
com.apple.clips	dict	
com.apple.iMovie	dict	
com.apple.mobilegarageband	dict	
com.apple.store.Jolly	dict	
com.burbn.instagram	dict	
com.facebook.Facebook	dict	
com.facebook.Messenger	dict	
com.fluntro.exif.lite	dict	
com.google.Gmail	dict	
com.mtr.mtrmobile	dict	
com.tencent.xin	dict	
com.yourappapp.revgram	dict	
net.whatsapp.WhatsApp	dict	
openrice.hongkong	dict	
org.whispersystems.signal	dict	
ph.telegra.Telegraph	dict	
us.zoom.videomeetings	dict	
Build Version	string	18C66
Device Name	string	iPhone
Display Name	string	iPhone
GUID	string	F4F83F4411B427F7DE46
ICCID	string	8985212210227055718
IMEI	string	352978115584444
IMEI 2	string	352978116182982
Installed Applications	array	
Last Backup Date	date	2022-10-17 11:30:45
MEID	string	35297811558444
Product Name	string	iPhone 11
Product Type	string	iPhone12,1
Product Version	string	14.3
Serial Number	string	DX3DJ00YN73T
Target Identifier	string	00008030-001155282E3
Target Type	string	Device
Unique Identifier	string	00008030-001155282E3
iTunes Files	dict	
iTunes Settings	dict	
iTunes Version	string	12.12.5.8

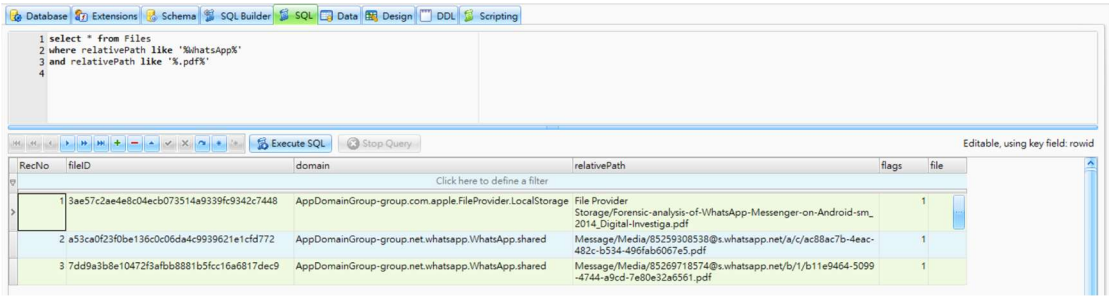
3. **<u>The phone user sent a pdf file using Whatsapp. The pdf file is a research paper with title: "Case study: From embedded system analysis to embedded system based investigator tools." What is the MD5 hash value of this file?</u>**

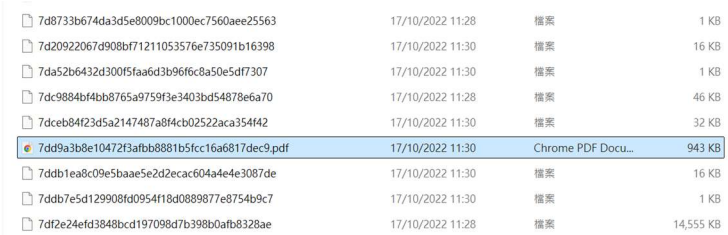
Answer: 7dd9a3b8e10472f3afbb8881b5fcc16a6817dec9

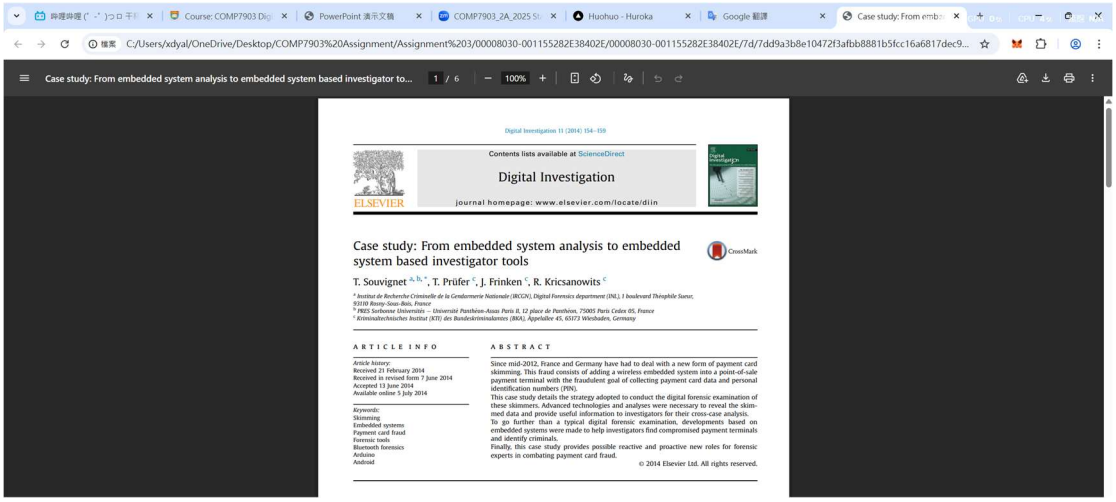
In the "Files" table, we can identify which WhatsApp-related files were previously shared. After applying a WHERE clause to search for both "Whatsapp" and "pdf", 3 PDF files were found. Using the fileID (the MD5 value), we can locate the PDF files in the corresponding folder path.

After opening all 3 PDF files, it is confirmed that 7dd9a3b8e10472f3afbb8881b5fcc16a6817dec9 is the research paper.

The screen capture is shown below:







4. <u>What is the electronic book in iBook APP?</u>

A: 红楼梦

B: 水浒传

C: 西游记

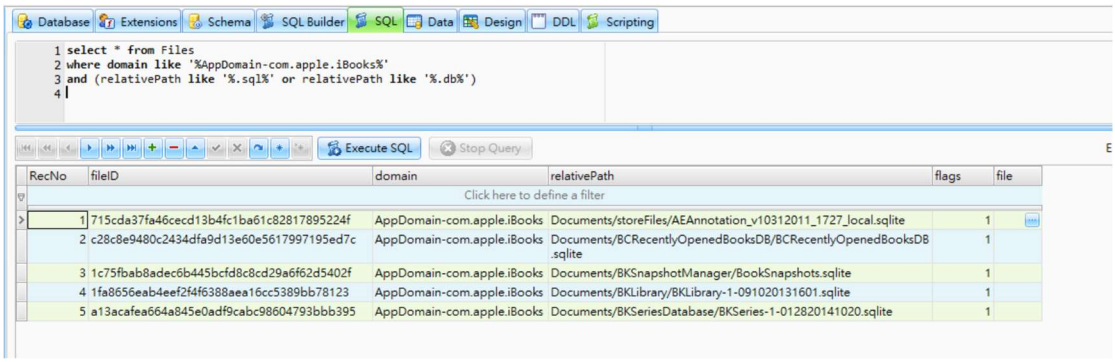
D: 三国演义

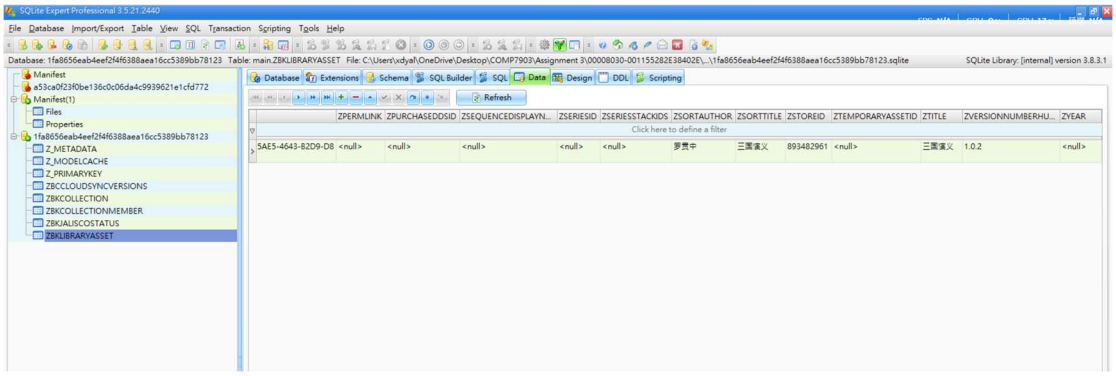
Answer: 三国演义

In the “Files” table, we can identify which .db file is related to the iBooks app.
From there, we can locate the “BKLibrary” SQLite database used by iBooks.

After opening this SQLite file using SQLite Expert, the book information can be found in the “ZBKLIBRARYASSET” table. In this table, “三国演义” appears as the only book asset.

The screen capture is shown below:





5. <u>Which part of the text did the user highlight in this e-book?</u>

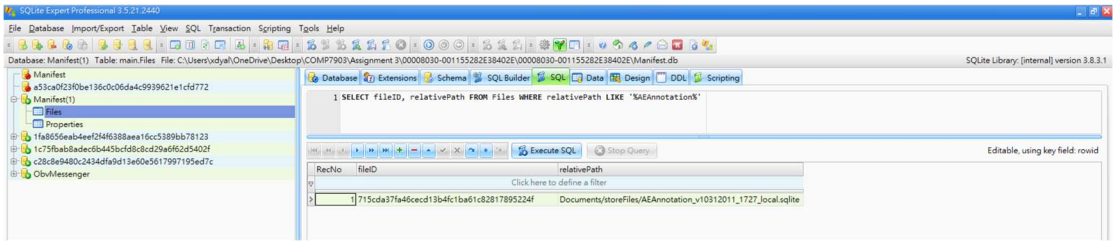
Answer: 卿有何妙计

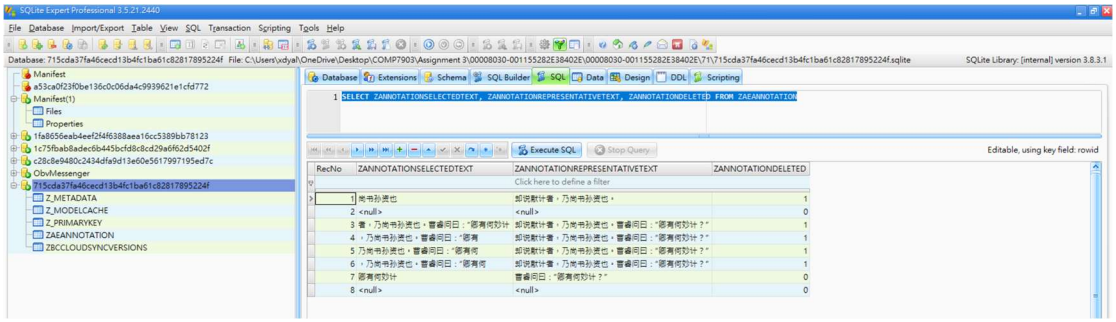
In the “Files” table, we can find the annotation .sqlite file named: 715cda37fa46cecd13b4fc1ba61c82817895224f.sqlite

After opening this .sqlite file, we can find the highlight information in the “ZAEANNOTATION” table.

There are several highlight annotations, but only the one with the text ”卿有何妙计” has ZANNOTATIONDELETED = 0. The other records have ZANNOTATIONDELETED = 1, which means they are deleted highlights.

The screen capture is shown below:

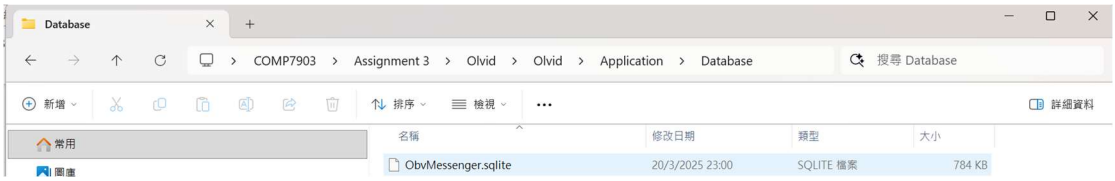




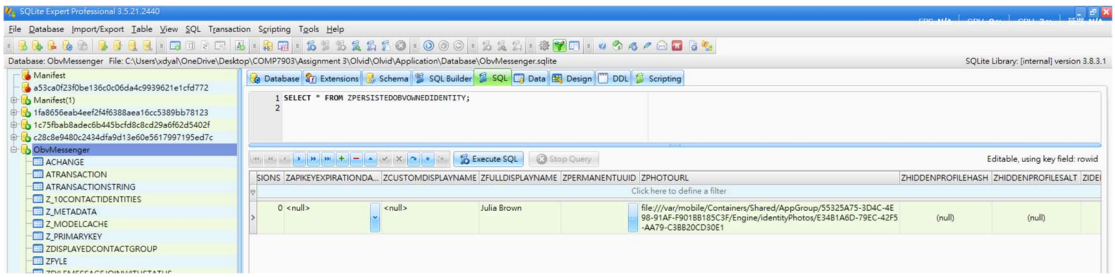
Case2: Please analyze the Olvid App Data “Olvid.zip” or “Olvid.rar”, and answer following questions with screen capture and your analysis detail (50’)

<u>Before answering the question, here is the background information for studying the “Olvid.zip”</u>

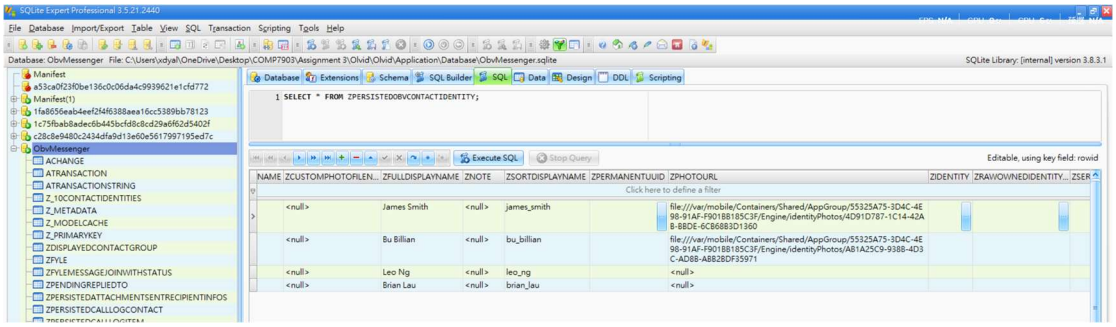
First, we can open the “ObvMessenger.sqlite” file to find out the device information.



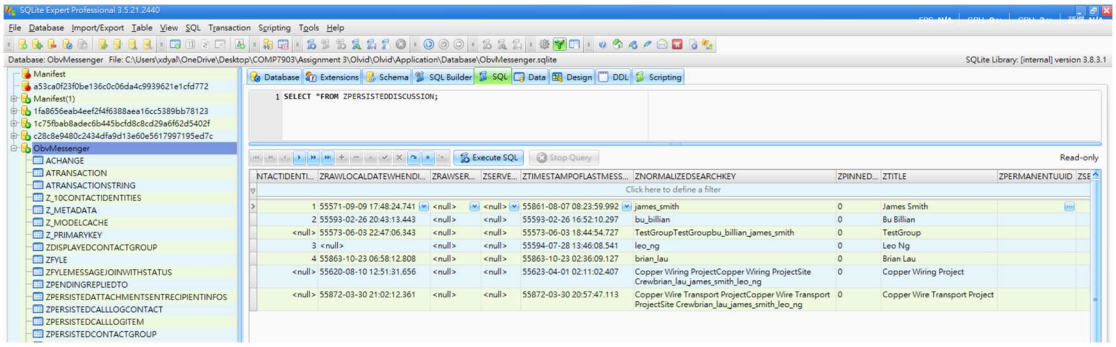
Second, we can find out that the owner is Julia Brown from the ZPERSISTEDOBVOWNEDIDENTITY.



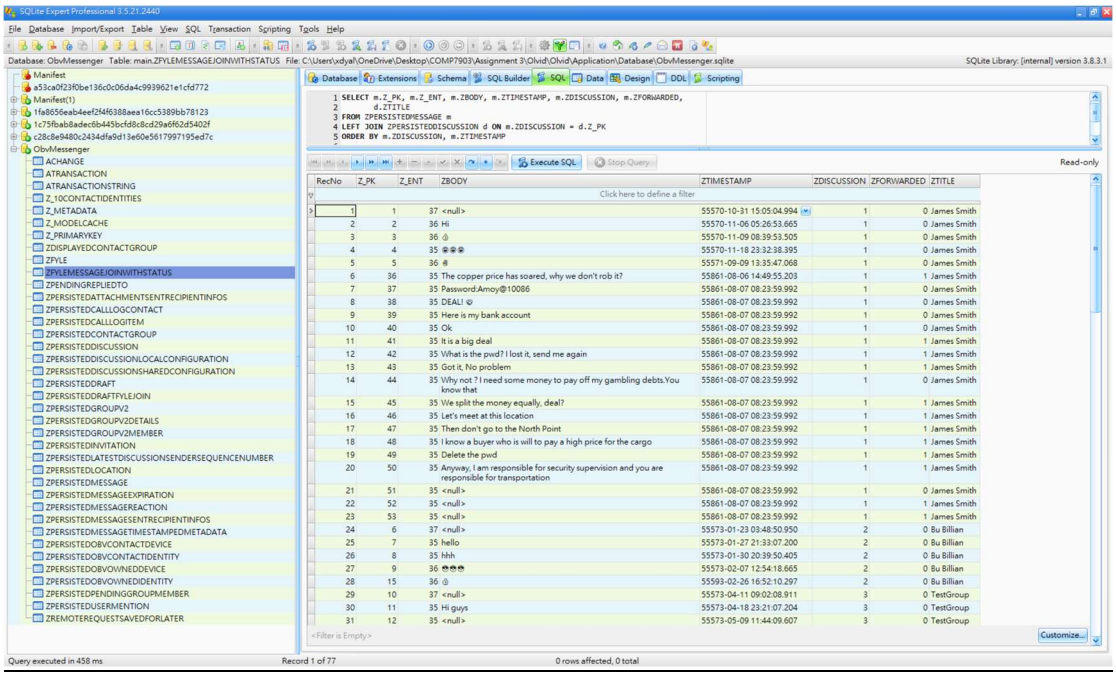
Third, there are 4 people (James Smith, Bu Billian, Leo Ng, Brian Lau) at the ZPERSISTEDOBVCONTACTIDENTITY.



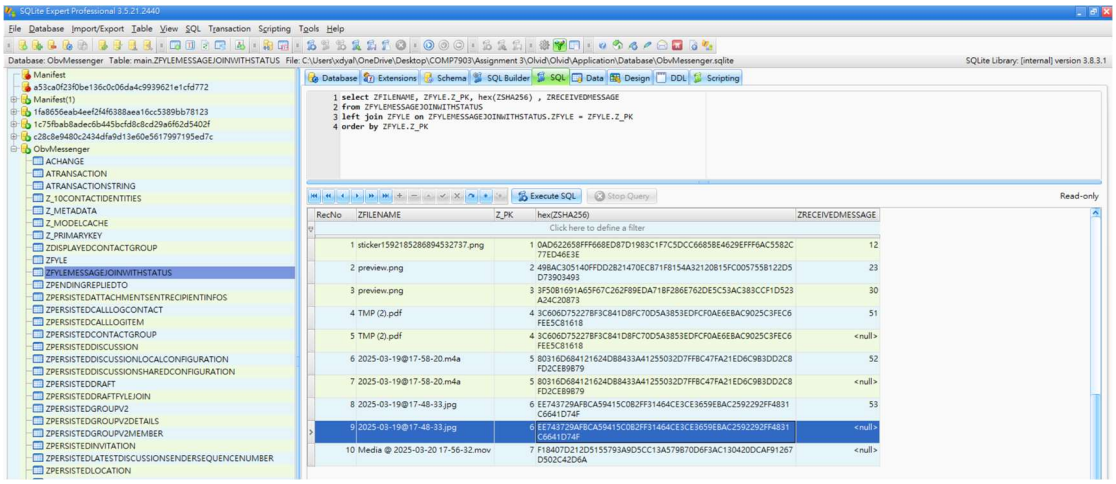
Fourth, we can find out about the discussion titles in the ZPERSISTEDDISCUSSION table. There are a total of 7 discussion sections.

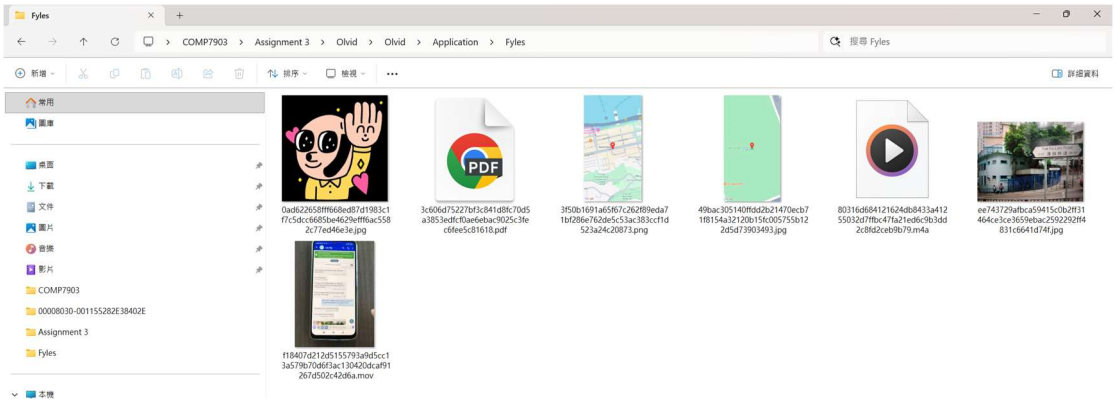


Fifth, we can study the discussion messages for different sessions in the ZPERSISTEDMESSAGE table by using the discussion ID (ZDISCUSSION).



Sixth, we can find out the actual file details from the ZFYLEMESSAGEJOINWITHSTATUS and ZFYLE.





1. Who is responsible for transportation?

Answer: James Smith

It is a complicated question that requires careful study of the conversion logic.
We can see that discussion sessions 1 and 5 contain related messages.

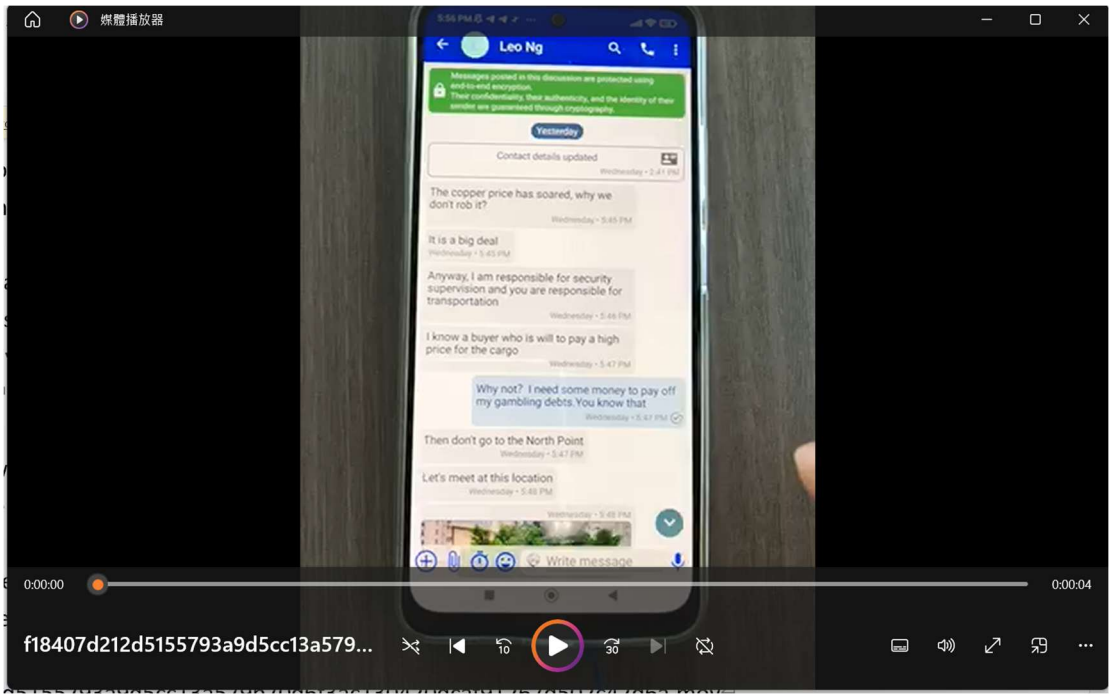
In discussion 5, we can clearly see that the device owner (Julia) is talking to Brian. The most important messages are as follows:

Julia: “Hi Brian, I found that Leo and James are suspicious. I have James’s phone in my hand and I can check his chat history.”
Brian: “Really!!!??? Can you forward the chat history to me?”
Julia: “Yes”

Therefore, we can understand that the reason some messages appear in both discussion 1 and 5 is because Julia had access to James’s phone history.

Next, we need to identify the sender and receiver in Session 1 messages. The evidence is in messages Z_PK 76 and Z_PK 77. Julia took a video of Leo and James conversion. After tracing the Z_PK 77 message (the video), we can find that it is the file:

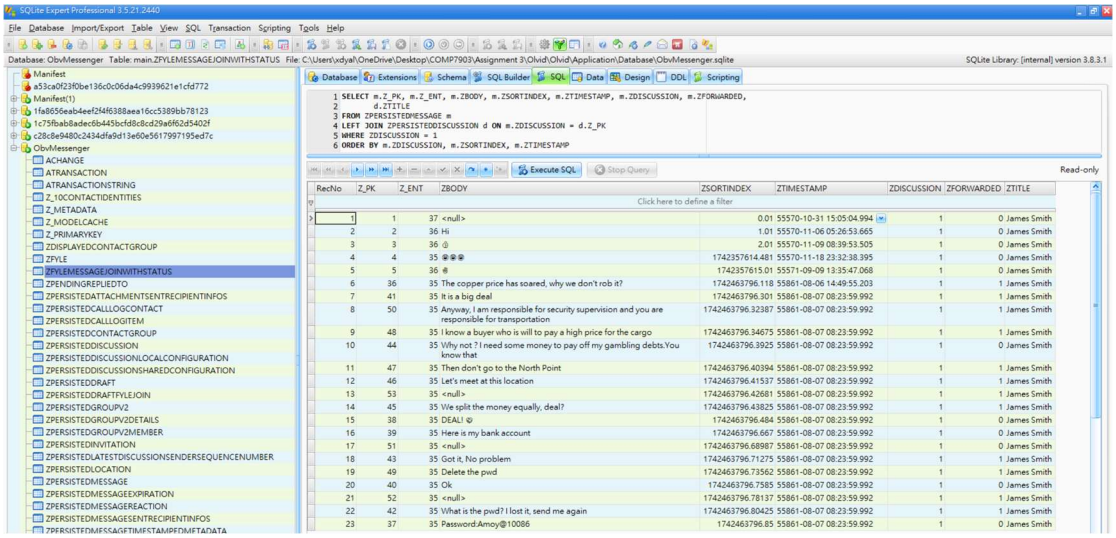
f18407d212d5155793a9d5cc13a579b70d6f3ac130420dcaf91267d502c42d6a.mov

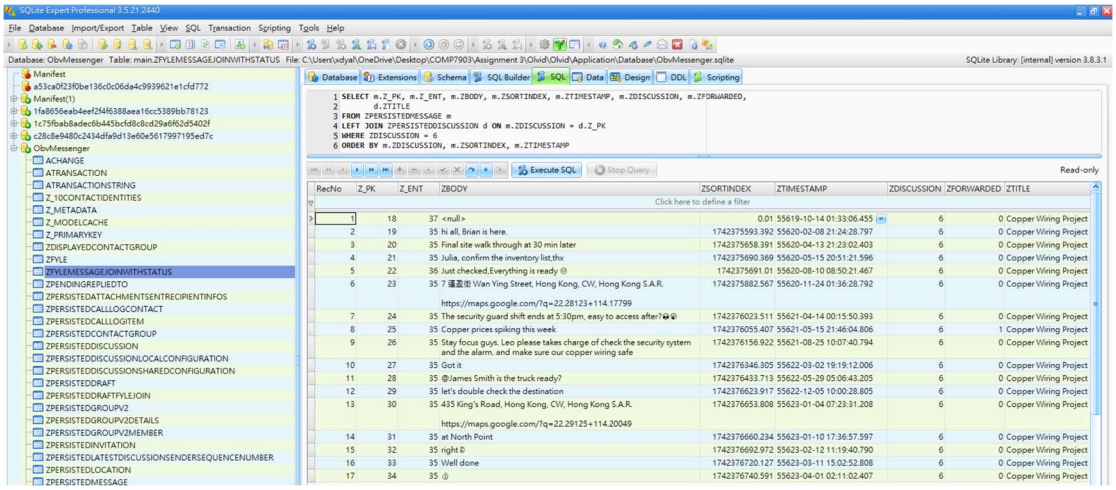


As a result, we can understand that discussion 1 is a one-to-one chat between Leo and James. Leo said, “Anyway, I am responsible for security supervision and you are responsible for transportation” (Z_PK = 50). This directly shows that James is responsible for transportation.

Moreover, in discussion 6, Brian said, “@James Smith is the truck ready?” (Z_PK = 28). This also shows that James is responsible for transportation.

The screen capture of discussion 1 and 6 is shown below:





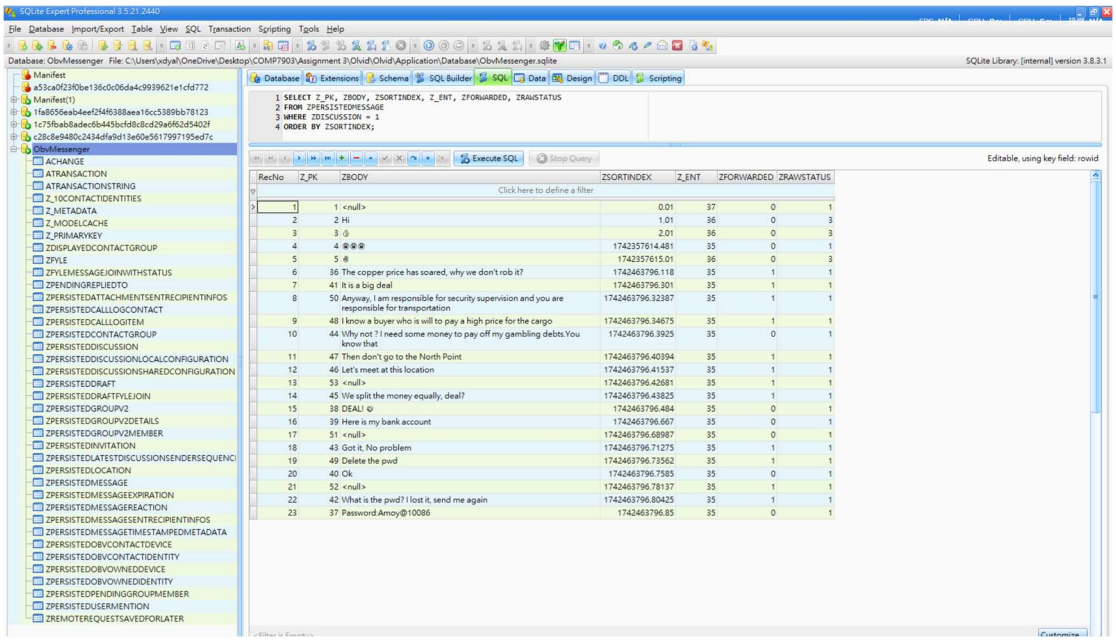
2. <u>What was James' motive for the robbery?</u>

Answer: James needed money to pay off his gambling debts.

In the previous study, we understand that discussion 1 is the conversation between James and Leo.

The relevant evidence is found in the row with Z_PK = 44, where James says:
"Why not? I need some money to pay off my gambling debts. You know that."

The screen capture is shown below:



3. When would Leo and James meet up?

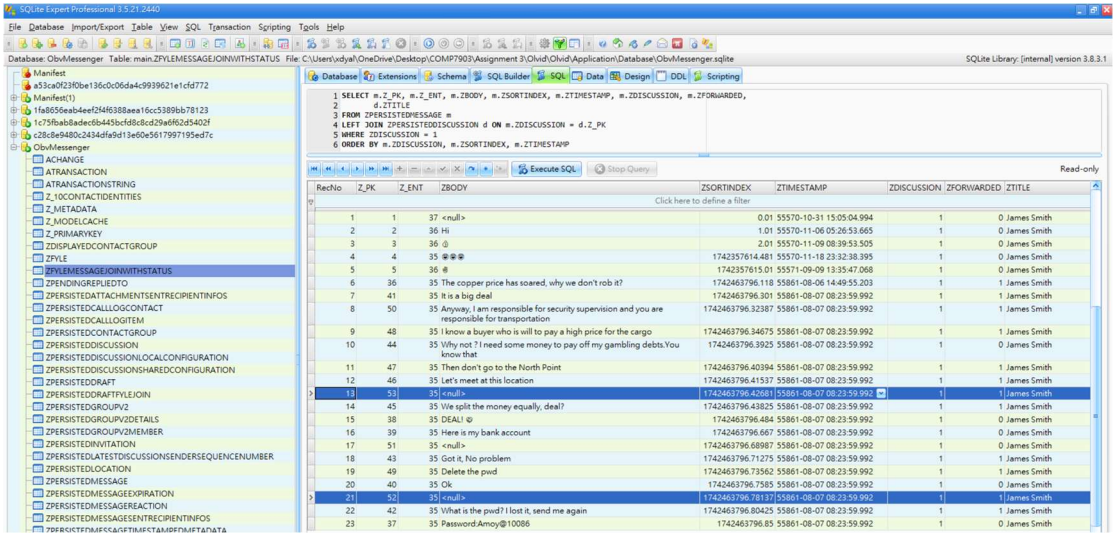
Answer: 7:00 p.m.

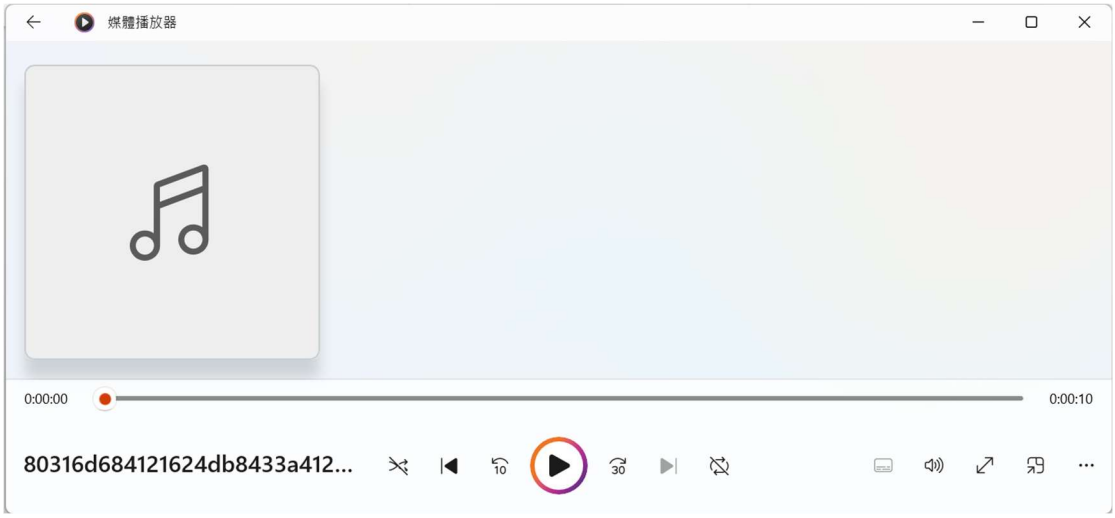
In Discussion 1, we can see that Leo and James planned to meet up. In message Z_PK 52, there is a media file. After tracing it through the ZFYLEMESSAGEJOINWITHSTATUS and ZFYLE tables, we can identify the sound recording file:

80316d684121624db8433a41255032d7ffbc47fa21ed6c9b3dd2c8fd2ceb9b79.m4a.

In the sound recording, it is clearly stated that they would meet at 7:00 p.m.

The screen capture is shown below:





4. <u>Where is the new destination for James?</u>

Answer: Pok Fu Lam Road

For this question, we need to study both Discussion 1 and Discussion 6.

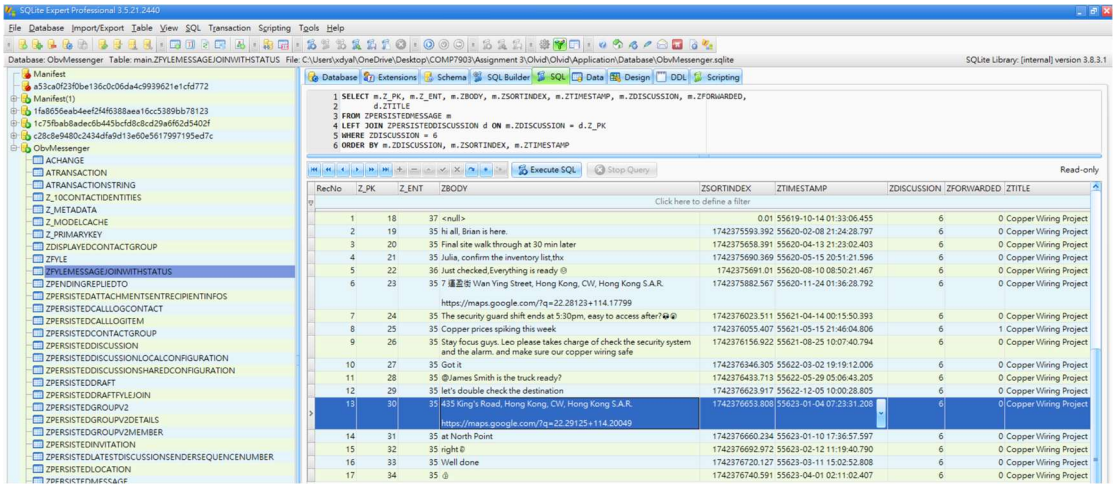
In Discussion 6, they planned to meet at North Point (435 King's Road, Hong Kong, CW, Hong Kong S.A.R.).

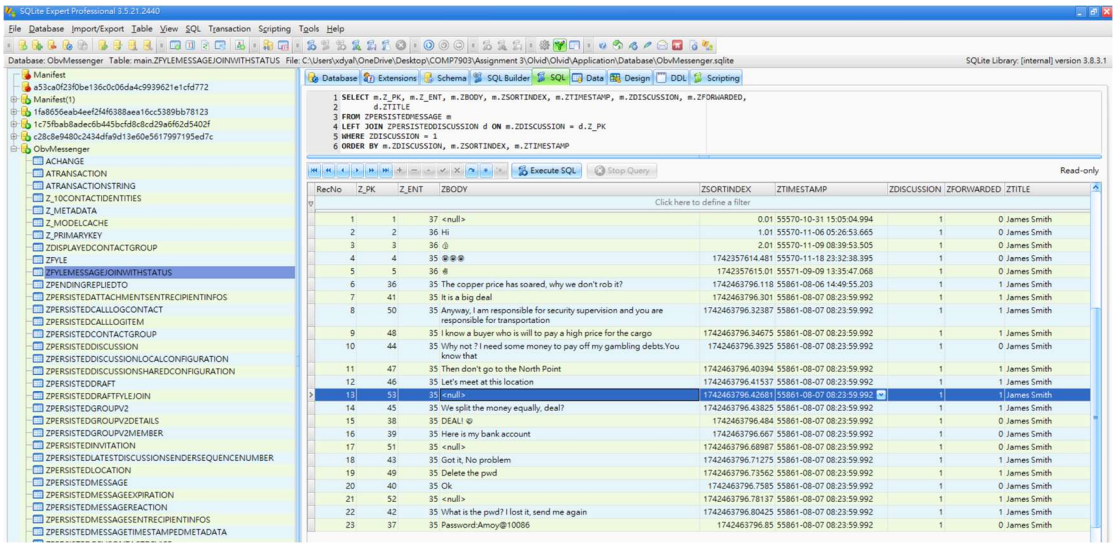
However, in Discussion 1, James discussed with Leo and sent the following messages:
“Then don't go to the North Point”
“Let's meet at this location”

After that, there is a photo attached to message Z_PK 53. After tracing this photo through the relevant tables, we can identify the file:
ee743729afbca59415c0b2ff31464ce3ce3659ebac2592292ff4831c6641d74f.jpg

It shows the new destination — Pok Fu Lam Road.

The screen capture of discussion 1 and 6 are shown below:



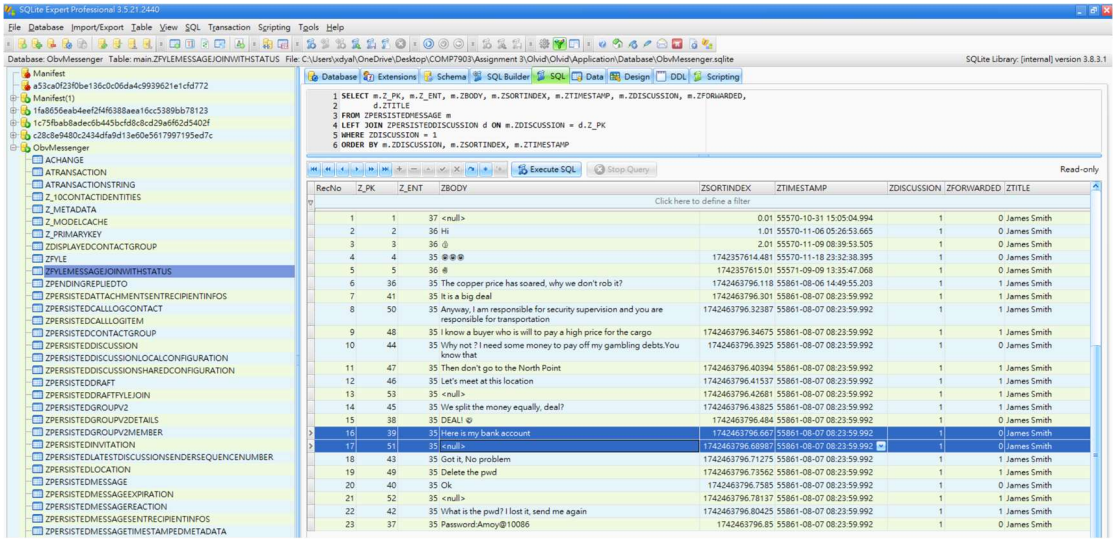
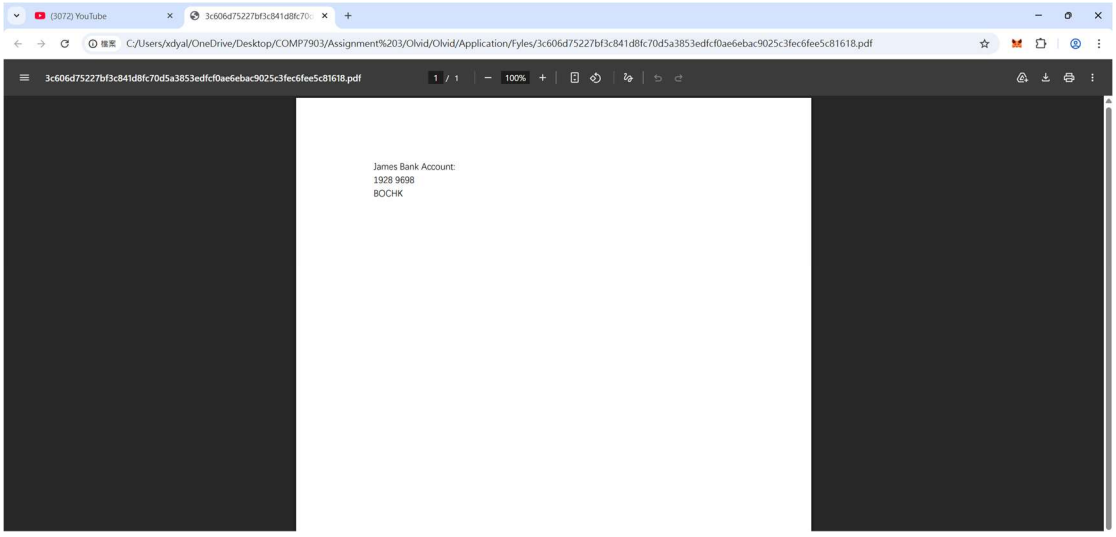


5. What is the bank account of James?

Answer: 1928 9698 BOCHK

In Discussion 1, James said, “Here is my bank account” (Z_PK 39), followed by a PDF attachment (Z_PK 51).

After tracing the Z_PK 51 attachment, we can identify the file -- TMP (2).pdf: 3c606d75227bf3c841d8fc70d5a3853edfcf0ae6ebac9025c3fec6fee5c81618.pdf.

To open this PDF, we can use the password shared in Discussion 1: “Amoy@10086” (Z_PK 37).

The screen capture is shown below: